Codeus Practice

HyperLogLog
Data Structure

About me

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- Software Engineer @ Intellias
- Co-organizer @ Codeus Community

HyperLogLog

probabilistic data structure
(typical error 1-2%)

estimates the number of distinct elements
(cardinality) in a very large collection of data.

memory efficient
(couple of kilobytes for billions of elements)

Problem

Count-distinct problem

Like “How many unique items are in this dataset?”

Count unique elements in distributed environment

Use Cases

Web & App Analytics

Estimate unique visitors, clicks, or active users without storing every ID. Ideal for real-time dashboards

Network Monitoring & Security

Track unique IPs or connections to detect DoS attacks and anomalies with minimal memory

Big Data Warehousing

Power APPROX_COUNT_DISTINCT in systems like Redshift, Snowflake, and BigQuery for fast distinct queries on billions of rows

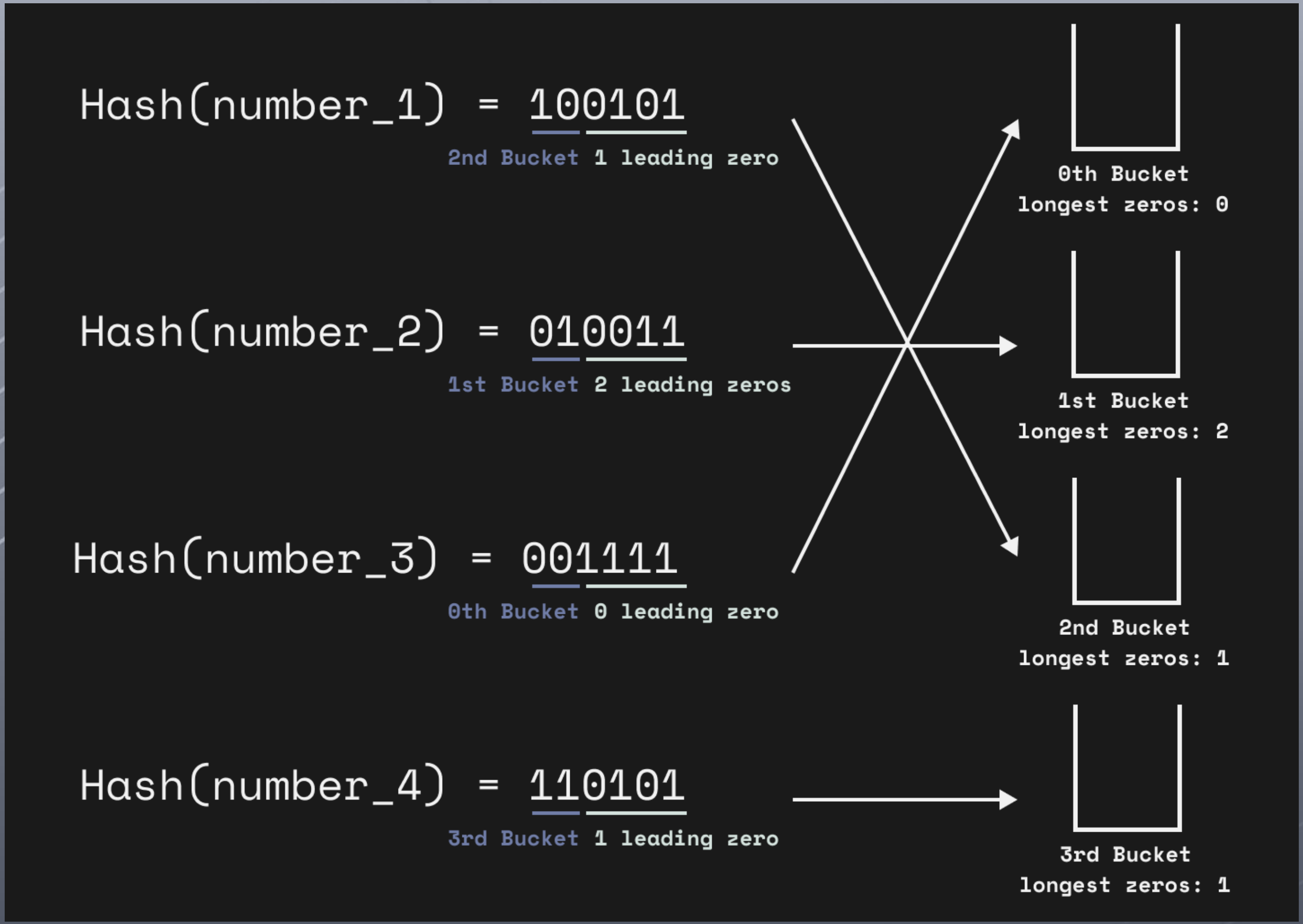
Streaming Analytics

Used in Flink/Spark to count unique events in sliding windows; mergeable across distributed nodes

Database & Search Engines

PostgreSQL (HLL extension), Redis (PFCOUNT), and Druid use HLL internally for efficient distinct counts and query optimization

How does it work?



Hash(number_5) = 100010
-> 2nd bucket and
overwrites 0-count to 2

Hash(number_6) = 010110
-> 1st bucket and
doesn't overwrite 0-count

https://towardsdatascience.com/hyperloglog-a-simple-but-powerful-algorithm-for-data-scientists-aed50fe47869/

LogLog?

$$\text{Cardinality}_{\text{LogLog}} = m \frac{2^{(L1+...+Lm)/m}}{\phi}$$

where m – number of hash functions,
L1, Lm - longest sequence of leading zeroes
ϕ - correction factor

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HyperLogLog

$$\text{Cardinality}_{\text{HyperLogLog}} = m\,\frac{\text{Harmonic Mean}(\,2^{L_i}\,)}{\phi}$$

where m – number of registers,
L1, Lm - longest sequence of leading zeroes
ϕ - correction factor

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Flashbang warning

Pros

Ultra Low Memory Footprint
(1.5 KB can encode counts beyond 1 billion with ~2% error)

Performance
(Adding an element is O(1), and merging two HLLs is O(m) which itself is fixed)

Distributed Use
(Data structure is designed to be merged)

Predictable Error Bound

Cons

Approximation
(Accuracy Trade-off)

No Item Storage or Lookup
(the actual data is not stored)

Limited Set Operations

Setup Upfront
(No Merging Across Different Parameters)

Redis: HyperLogLog

Redis

In-memory NoSQL key-value storage

Support multiple data structures:

- Json
- Sets
- HyperLogLog
- BloomFilter
- more

Main commands

PFADD

Adds elements to a HyperLogLog key.
Creates the key if it doesn't exist.

https://redis.io/docs/latest/commands/?group=hyperloglog

Main commands

PFADD

PFCOUNT

Returns the approximated cardinality of the set(s) observed by the HyperLogLog key(s).

https://redis.io/docs/latest/commands/?group=hyperloglog

Main commands

PFADD

PFCOUNT

PFMERGE

Merges one or more HyperLogLog values into a single key.

https://redis.io/docs/latest/commands/?group=hyperloglog

Spring Redis

Spring Redis

```java
@Configuration
public class RedisConfig {
  @Bean
  public RedisTemplate<String, String> redisTemplate(
    RedisConnectionFactory connectionFactory) {
    RedisTemplate<String, String> template = new RedisTemplate<>();
    template.setConnectionFactory(connectionFactory);
    template.setKeySerializer(new StringRedisSerializer());
    template.setValueSerializer(new StringRedisSerializer());
    return template;
  }
}
```

https://docs.spring.io/spring-data/redis/reference/redis/template.html

Spring Redis

```
> PFADD bikes Hyperion Deimos Phoebe Quaoar
(integer) 1
> PFCOUNT bikes
(integer) 4
> PFADD commuter_bikes Salacia Mimas Quaoar
(integer) 1
> PFMERGE all_bikes bikes commuter_bikes
OK
> PFCOUNT all_bikes
(integer) 6
```

```java
public void redisHyperLogLogOperations(RedisTemplate<String, String> redis) {
  redis.opsForHyperLogLog().add("bikes", "Hyperion", "Deimos", "Phoebe", "Quaoar");
  redis.opsForHyperLogLog().size("bikes"); // 4

  redis.opsForHyperLogLog().add("commuter_bikes", "Salacia", "Mimas", "Quaoar");

  redis.opsForHyperLogLog().union("all_bikes", "bikes", "commuter_bikes");
  redis.opsForHyperLogLog().size("bikes"); // 6
}
```

Thank you

- Author: Serhii Kravchuk
- My LinkedIn: [Link]
- Date: September 2025
- Join Codeus community in Discord
- Join Codeus community in LinkedIn